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Subtropical forests exhibit contrasting resistance and resilience to unexpected extreme ice storms

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Abstract

Background Extreme ice storms (EIS) can severely disrupt forest productivity and stability, yet their differential impacts across forest types remain poorly understood, particularly in subtropical regions. Here, we assessed changes in gross primary productivity (GPP), resistance (Rt), and resilience (Rs) across evergreen needle-leaved forest (ENF), evergreen broad-leaved forest (EBF), deciduous broad-leaved forest (DBF), and mixed forest (MF) in mid-to-north subtropical China following two unexpected EIS events in early 2008 and 2018. Random Forest and Partial Least Squares Structural Equation Modeling (PLS-SEM) were used to identify key drivers and disentangle their underlying causal pathways.

Results Except for DBF, both EIS events led to declines in forest GPP, with the stronger 2008 event exerting a more pronounced suppressive effect. Forest stability exhibited a clear Rt–Rs trade-off across forest types—the 2008 event was characterized by lower Rt but higher Rs, whereas the opposite pattern was observed in 2018. Among forest types, DBF showed the highest stability, while EBF exhibited the lowest. Freezing duration had contrasting effects between the two events: prolonged freezing duration significantly reduced Rt and Rs in 2008, but enhanced both in 2018. Moreover, freezing intensity, particularly minimum-temperature stress and cumulative precipitation, was the foremost determinant of forest stability, whereas geographical factors acted mainly indirectly through climate interannual variability, freezing intensity, and phenology. Furthermore, biological attributes made relatively modest overall contributions but provided important insights into forest-type-specific stability mechanisms. Specifically, phenology dominated DBF stability, whereas forest structure acted primarily through leaf area index in DBF and MF and through stand age in ENF and EBF.

Conclusions Subtropical forest responses to EIS are highly heterogeneous and jointly controlled by disturbance intensity, forest composition, and spatial environmental context. By revealing the mechanisms underlying forest-type-specific stability responses to EIS, this study provides useful insights for differentiated risk-resilience management aimed at enhancing ecosystem stability and sustainable development.

Keywords Extreme ice storm, Subtropical forest, Gross primary productivity, Forest stability, Resistance, Resilience

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Introduction

Climate change has significantly intensified the frequency, intensity, and duration of extreme climate events, including ice storms, droughts, and heat waves (Piao et al. 2019; Li et al. 2021a, b). Ice storms, which predominantly occur in mid-to-high latitude regions, are characterized by snowfall or rainfall accompanied by abrupt and prolonged decreases in temperature (Lemon 1961; Lafon and Speer 2002; Hellmer et al. 2015). Over the past few decades, an alarming increase in the occurrence of extreme ice storms (EIS) has been observed in mid-to-low latitude regions due to climate change (Klima and Morgan 2015; Deschênes et al. 2019), negatively affecting the structure and function of terrestrial ecosystems (Xu et al. 2013; Hellmer et al. 2015), with cascading socio-economic consequences arising from disruptions to ecosystem services, infrastructure systems, and regional economic activities (Zhou et al. 2011). Given that the ongoing climate change is projected to increase the frequency and intensity of EIS and alter the range where they occur (Guo et al. 2023; Yang et al. 2026), it is essential to understand how different forest types respond to such events.

Forests play crucial roles in regulating global carbon and water cycles (Huang et al. 2020). Gross primary productivity (GPP), a key parameter characterizing ecosystem functions and processes (Gilabert et al. 2015), has been shown to be more sensitive to extreme climate events than to gradual climate change (Hanson et al. 2006). For instance, late spring frost not only directly suppresses vegetation photosynthesis, resulting in an immediate decline in GPP, but also increases the thermal requirement threshold for the initiation of the subsequent growing season (Rubio-Cuadrado et al. 2021; Wang et al. 2025a). This delay in the start of the growing season causes a phenological shift, leading to a prolonged reduction in GPP in the following year by shortening the functionally active growing period (Wang et al. 2025a). Similarly, extreme cold events expose plants to temperatures exceeding their tolerance thresholds, resulting in tissue damage, impaired photosynthetic function, and increased mortality risk, all of which contribute to a substantial decline in GPP (Körner 2016; Yuan et al. 2021; Yang et al. 2026). However, due to the low frequency of EIS, current research predominantly focuses on evaluating the extent of mechanical damage to forests at limited spatial scales (Xu et al. 2013; Zhou et al. 2017; Lu et al. 2020), a comprehensive and comparative evaluation of the physiological responses of different forest types across broader spatial scales is still lacking. Such climate-driven physiological shifts not only intensify spatiotemporal variability in the forest carbon cycle, but also compromise carbon stock durability, thereby undermining the stability of regional socio-economic activities. These uncertainties further challenge the reliability of existing carbon credit accounting frameworks, leading to a significant disconnect between actual emission reduction effectiveness and economic returns (Anderegg et al. 2025). Therefore, investigating the responses of various forest types to EIS through the lens of forest phenology and its linkage with GPP can enhance our understanding of how EIS affect forest dynamics and ecosystem functioning, thereby informing sustainable forest management and climate adaptation strategies.

The capacity of an ecosystem to maintain its health and productivity constitutes a central question in ecology, and is commonly referred to as ecosystem stability (Pimm 1984). Currently, ecosystem stability metrics have evolved from purely ecological concepts into key decision-support tools for resource management and climate governance, directly informing forest management optimization and ecosystem service assessment (Li et al. 2025). Among the metrics used to quantify this capacity, resistance and resilience have emerged as key concepts for understanding ecosystem dynamics under a changing climate (Pimm 1984; Wang et al. 2024b). Previous studies have revealed that GPP can effectively reflect the functional recovery of ecosystems following extreme events (Frazier et al. 2013; Zhang et al. 2025). Given the high sensitivity and widespread use of GPP as a proxy for vegetation response to extreme events, resistance and resilience metrics derived from GPP can be effectively employed to assess forest growth responses to EIS (Yuan et al. 2021; Yang et al. 2026). Recent studies have demonstrated that satellite-based solar-induced chlorophyll fluorescence (SIF) data provides more accurate estimations of GPP compared to traditional methods (Qiu et al. 2022; Hao et al. 2021; Shekhar et al. 2022), enabling more precise characterization of forest dynamic changes and providing critical technical support for precision forestry (Wang et al. 2025b), thereby facilitating the large-scale assessment of forest resistance and resilience under environmental stress.

The magnitude and extent of forest destruction caused by EIS are influenced by a range of factors, including forest structure (e.g., species composition and forest age), freezing intensity, and site conditions (Deschênes et al. 2019; Lu et al. 2020; Xiang et al. 2025), which subsequently affect forest resistance and resilience. For example, Lisella et al. (2025) revealed that Turkey oak exhibited lower resistance than silver fir to late frost in drier site conditions but showed greater resistance in wetter ones. Vitasse et al. (2019) reported that deciduous broad-leaved species (oak and beech) are more sensitive to late frost, which strongly reduces current year growth but with higher growth resilience when compared to coniferous species (silver fir, Norway spruce and

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European larch). Forest crown damage intensified with the severity of ice storm (Nagel et al. 2016), resulting in greater adverse impacts on overstory trees in evergreen broad-leaved forests compared to those in deciduous broad-leaved forests in subtropical regions (Lu et al. 2020). Meanwhile, forest age also influences damage and recovery of forests impacted by EIS, as young trees were more susceptible but exhibited higher resistance to EIS than other trees due to sheltering by large trees (Tavankar et al. 2019). Resistance of forest trees to ice damage is an emergent consequence of tree attributes, species origin, site conditions, and human disturbance (Zhou et al. 2017). For instance, forests at higher elevations generally experience greater damage than those at lower elevations (Chen and Sun 2010), and slope angle, rather than slope aspect, is negatively associated with the total damage caused by EIS (Xiang et al. 2025). However, EIS can also yield potential beneficial effects, such as providing necessary moisture for forest growth (Cao et al. 2020) and increasing sunlight availability for understory vegetation due to the destruction of the upper canopy layer (Lafon 2016). These changes alleviate competitive pressure among tree species for light resources, thereby enhancing forest species diversity (Lafon 2004) and contributing to forest recovery, e.g., forests affected by ice storms can recover fully within 2 years (Vitasse et al. 2019; D'Andrea et al. 2021). Although the heterogeneous patterns of forest damage can be attributed to a combination of freezing intensity, site conditions, and species-specific sensitivity to disturbances (Nagel et al. 2016), the relative importance of these factors on the resistance and resilience of different forest types remain poorly understood, particularly in subtropical regions.

Southern China is generally characterized by a warm and moist subtropical climate, with EIS occurring relatively infrequently. However, influenced by the La Niña phenomenon and abnormal atmospheric circulation, two significant EIS events occurred in early 2008 and 2018, respectively (Zhou et al. 2011; Qin and Li 2020). The 2008 EIS, in particular, was the most extensive, prolonged, and intense meteorological event over the past 50 years (Wang et al. 2020), resulting in severe vegetation damage and significant changes in ecological processes (Zhao et al. 2020). Moreover, the event triggered substantial socio-economic disruptions and systemic failures in critical energy infrastructure, exerting profound impacts on regional economic development (Zhou et al. 2011). To date, field studies at large spatial scale on the responses of different forest types to EIS remain limited. These two unexpected EIS events provide a unique opportunity to investigate response differences among forest types and the underlying regulatory mechanisms. Therefore, this study specifically aims to address: (1) how does GPP respond to EIS across different forest types? and (2) what are the key factors regulating resistance and resilience, and how do these factors influence forest responses to EIS? The findings of this study are expected to support forest management practices, enhance ecosystem resilience, and inform climate adaptation strategies under increasing climate extremes.

Materials and methods

Study region

The study region encompasses five provinces in southern China—Hunan, Jiangxi, Hubei, Anhui, and Henan (24° 52′–36° 16′ N, 108° 46′–118° 44′ E)—covering a geographical gradient from mid-to-north subtropical China to the warm-temperate transition zone (Fig. 1). Characterized by a typical monsoon climate with four distinct seasons, the region has a mean annual temperature (MAT) ranging from 15 to 18 °C and mean annual precipitation (MAP) between 1000 and 1800 mm. The predominant forests in this region include evergreen needle-leaved forest (ENF), evergreen broad-leaved forest (EBF), deciduous broad-leaved forest (DBF), and mixed forest (MF) composed with coniferous and broad-leaved trees. In early 2008 and 2018, two unexpected EIS events occurred in southern China (hereafter referred to as the 2008 and 2018 events, respectively). These events caused severe damage to terrestrial ecosystems, with the 2008 event being substantially more intense than the 2018 event (Qin and Li 2020; Wang et al. 2020).

Data sources

Solar-induced chlorophyll fluorescence (SIF) is distinguished by its unique capability to directly monitor actual photosynthetic activity, thereby providing an effective means for assessing vegetation photosynthesis (Zhang et al. 2022, 2023). Consequently, we utilized the Global 'OCO-2' SIF dataset (GOSIF), which spans from 2003 to 2022 and with a spatial resolution of 0.05° and a temporal resolution of 8 days, to extract phenological metrics of forests (http://data.globalecology.unh.edu/data/GOSIF_v2/). Additionally, we employed the global annual GPP dataset (GOSIF-GPP), derived through the integration of GOSIF data and inverted flux tower observations (http://data.globalecology.unh.edu/data/GOSIF-GPP_v2/), covering the period from 2003 to 2022 with a spatial resolution of 0.05°, to analyze trends of forest annual mean GPP variations. Both dataset, GOSIF and GOSIF-GPP, have demonstrated strong consistency with ground-based observation data (Li and Xiao 2019), and have been widely applied in studies on vegetation activity dynamics and global C cycle (Qiu et al. 2022; Shekhar et al. 2022).

Meteorological data were obtained from the National Meteorological Information Center (https://data.cma.

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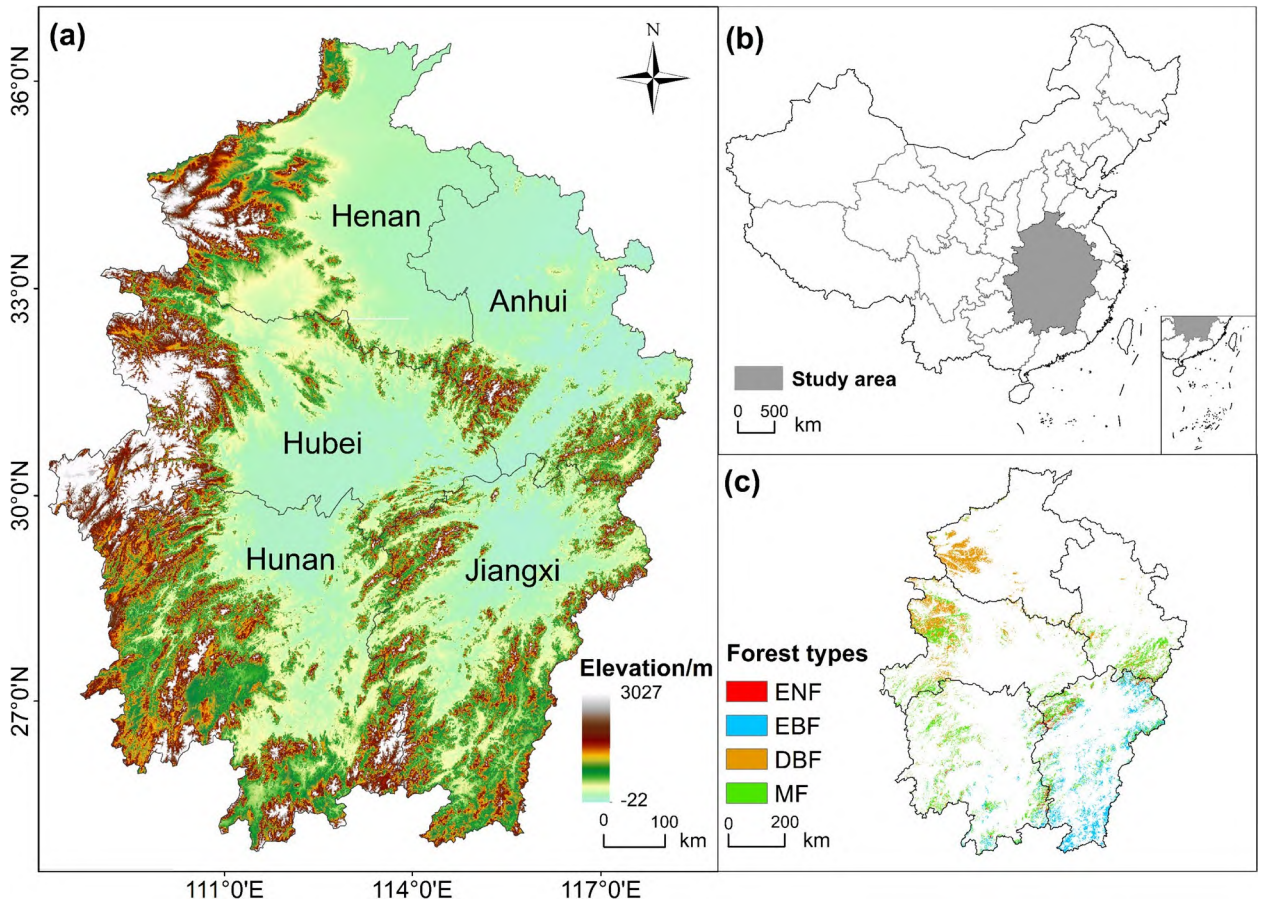


Fig. 1 The location of study region: **a** Topography of the study region; **b** Geographical location of the study region; **c** Spatial distribution of four types of forest

cn/). The daily datasets included precipitation, mean temperature and minimum temperature for January and February of both 2008 and 2018, which were utilized to determine frozen days and intensity. Climate Interannual Variability (IAV) is a critical metric for assessing climatic stability, quantified through the coefficient of variation in climate data (Meng et al. 2021). This index was calculated using monthly climate data spanning from 1978 to 2007, encompassing three key climatic variables: monthly mean temperature, the minimum temperature, and monthly precipitation. Specifically, the annual data were first synthesized from the monthly data, and the coefficient of variation was then computed as the ratio of the standard deviation of the annual data over the 30-year period to its corresponding mean. Meteorological station data were interpolated into raster data with a spatial resolution of 1 km using Anusplin (v.4.2), employing the TVPTPS2 model along with a three-variable partial thin plate smoothing splines function interpolation method. Elevation was used as a covariate during this process, while a spline number of 2 and a smoothing method that minimized the generalized cross-validation (GCV) score were applied.

Forest types were derived from MCD12Q1 (version 6) (https://lpdaac.usgs.gov/products/mcd12q1v006/), which provide global annual land cover type data at a resolution of 500 m. To minimize the influences of land use changes caused by anthropogenic activities, our analysis exclusively focused on pixels with consistent land cover classifications throughout the study period (from 2003 to 2022). Forest structure variables included forest age and leaf area index (LAI). Forest age was derived from a 30-m resolution annual forest age dataset for China covering the period 1986–2022 (https://doi.org/10.6084/m9.figshare.24464170) (Shang et al. 2025). To establish a pre-disturbance baseline for the ice storms that occurred in early 2008 and 2018 (January–February), we used forest age data from 2007 and 2017, respectively. This selection accurately captures the cumulative growth prior to each event and the onset of the new growing season, thereby minimizing uncertainties associated with storm-induced structural changes. LAI data were acquired from

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the MODIS 500 m resolution 8-day composite product (MOD15A2H) (https://lpdaac.usgs.gov/products/mod15a2hv006/), and the mean LAI values for the period from January to February in 2008 and 2018 were calculated.

Geographical variables (e.g., elevation, slope, and aspect) were extracted from the SRTM 90 m Digital Elevation Database (DEM) (https://srtm.csi.cgiar.org/) (Jarvis et al. 2008) by ArcGIS (v.10.2). The geographical variables also included longitude and latitude. The soil data, such as bulk density, porosity, and pH, were extracted from the China dataset of soil properties for land surface modeling (http://globalchange.bnu.edu.cn/research/soil2) (Shangguan et al. 2013).

All the spatial grid data were resampled to a uniform resolution of 1 km, if required, utilizing the kriging interpolation algorithm.

Methods

Freezing intensity determination

To comprehensively evaluate the impact of EIS, freezing intensity was quantified using multiple indicators, including freezing duration (FD), cumulative precipitation (Pcum), and absolute minimum temperature (Tminabs). FD was defined as the total number of days with daily precipitation > 0 mm and daily mean temperature ≤ 1 °C, following the criteria of Wang et al. (2008). The spatial distribution and frequency statistics of FD for the two EIS events are shown in Fig. S1. Pcum was calculated as the sum of daily precipitation during FD, serving as a proxy for ice accumulation and associated mechanical loading. Tminabs was defined as the lowest daily minimum temperature for each pixel during FD, representing peak physiological cold stress. Daily meteorological data (precipitation, mean temperature, and minimum temperature) for January and February of 2008 and 2018 were interpolated into 1-km resolution raster images using the Anusplin (v.4.2) package. These daily rasters were then overlaid to generate the spatial datasets for FD, Pcum, and Tminabs. This multi-indicator framework for characterizing ice storm severity is consistent with previous studies (Gao 2009; Wang et al. 2020; Feng et al. 2021) and provides a more comprehensive quantification of freezing intensity.

Forest phenology

Here, we identify the two main vegetation phenology parameters—the start-of-growing season (SOS) and the end-of-growing season (EOS)—at the pixel scale within the study area. A total of 46 SIF images per year was utilized for time series reconstruction from 2003 to 2022. The Savitzky-Golay filtering method, configured with a polynomial number of 3 and a sliding window size of 9, was applied to smooth the SIF time series curves (Chen et al. 2004). Subsequently, a straightforward yet effective dynamic threshold approach was utilized to extract forest phenological metrics from the reconstructed SIF time series using the following equations:

$$SOS = SIF_{\min 1} + (SIF_{\max} - SIF_{\min 1}) \times R_{\text{thd}} \quad (1)$$

$$EOS = SIF_{\min 2} + (SIF_{\max} - SIF_{\min 2}) \times R_{\text{thd}} \quad (2)$$

where SIF_{max} is the maximum SIF value in a year, SIF_{min1} is the minimum value in the range from the first day of the year to the day with SIF_{max}, SIF_{min2} is the minimum value in the range of SIF_{max} to the rest day of the year, and R_{thd} is the threshold value. Here we selected 0.2 as the threshold for phenology extraction according to Liu et al. (2023).

Forest resistance and resilience

Resistance (Rt) reflects the extent to which forest growth is maintained during EIS compared to the pre-ice storm state, representing the ability of forests to alleviate stress and sustain normal growth amidst disasters. Resilience (Rs) indicates the proportion of recovery towards the pre-ice storm state after experiencing EIS, signifying the capacity of forests to restore themselves back to their normal states following disturbances. Rt and Rs for the four types of forest in relation to EIS were calculated using a method proposed by Yi and Jackson (2021), involving pixel-by-pixel ratio of GPP values before and after the events:

$$Rt = \frac{GPP_{dist}}{\overline{GPP_{pre}}} \quad (3)$$

$$Rs = \frac{GPP_{post}}{\overline{GPP_{pre}}} \quad (4)$$

where $\overline{GPP_{pre}}$ and GPP_{post} are the average annual GPP values two years before and after the ice storm year, respectively. GPP_{dist} is the annual GPP value in ice storm year. To minimize the confounding effects of annual variations in GPP value resulting from climate warming and afforestation, the two years preceding the ice storm were selected as the reference period to represent normal conditions (White et al. 2017), since previous studies have demonstrated that vegetation affected by ice storms can recover its normal growth within 2 years (Vitasse et al. 2019; D'Andrea et al. 2021).

Key factor identification

To identify the key factors driving the variability in forest Rt and Rs under EIS, we initially selected 22 environmental predictors spanning multiple ecological dimensions (Table 1). To minimize the influence of multicollinearity, we calculated the Variance Inflation Factor (VIF) for

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Table 1 Environmental variables across ecological dimensions used to analyze forest Rt and Rs under EIS

Category	Predictor variable	Abbreviation	Units
Climate IAV	Mean temperature IAV	MeanTem	–
	Minimum temperature IAV	MinTem	–
	Precipitation IAV	Pre	–
Forest structure	Forest age	Age	year
	Leaf area index	LAI	m^2/m^2
Freezing intensity	Freezing duration	FD	day
	Cumulative precipitation during freezing	Pcum	mm
	Absolute minimum temperature during freezing	Tminabs	°C
Geography	Longitude	Long	°
	Latitude	Lat	°
	Elevation	Elev	m
	Slope	Slop	°
	Aspect	Aspe	°
Phenology	Start of growing season	SOS	day of year (DOY)
	End of growing season	EOS	day of year (DOY)
Soil properties	Soil bulk density	BD	g/cm^3
	Soil pH	PH	–
	Soil porosity	POR	%
	Soil organic matter	SOM	g/kg
	Total nitrogen	TN	g/kg
	Total phosphorus	TP	g/kg
	Total potassium	TK	g/kg

all environmental predictors. We adopted a maximum VIF threshold of 10, a criterion widely recognized in ecological research to ensure model robustness while retaining essential ecological information. According to comprehensive simulation studies (Dormann et al. 2013), maintaining VIF values below 10 ensures that both machine-learning algorithms (e.g., Random Forest) and latent variable methods (e.g., PLS-SEM) can provide reliable variable importance assessments and stable path coefficients without being severely distorted by inter-predictor correlations. All variables retained in the final models satisfied this criterion, as shown in Supplementary Table S1.

Subsequently, we developed RF regression model using the scikit-learn package (v. 1.4.2) in Python (v. 3.11.0) to quantify the contributions of various variables to Rt and Rs across different forest types. The Random Forest (RF) model is an ensemble learning model that generates numerous decision trees and combines them to form a more robust predictive model (Breiman 2001). This approach improves prediction accuracy by averaging the results from numerous decision trees, each trained on randomly selected subsets of data and variables. Compared with conventional statistical methods, RF better captures complex interactions among predictor variables and handles highly collinear predictors more effectively (Cutler et al. 2007). Model conducted via the RandomForestRegressor function, and variable importance was assessed using the permutation importance method.

The dataset was randomly split into training and independent testing sets at a 3:1 ratio (random_state = 42). Random forest hyperparameters were optimized via grid search with tenfold cross-validation, focusing on two parameters: the maximum number of features for node splitting (max_features, even integers 2–20) and the minimum number of samples at leaf nodes (min_samples_leaf, integers 1–10). The number of trees (n_estimators) was fixed at 500 based on preliminary experiments (Breiman 2001). The optimal parameter set was selected by minimizing Mean Squared Error (MSE) during cross-validation (Table S2). Model performance was evaluated using the coefficient of determination (R^2), Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), which collectively represent the model's explanatory power, average bias, and sensitivity to extreme errors, respectively.

To assess variable importance, permutation importance was applied instead of Gini importance to avoid feature scale bias. Using the test set, 100 permutations were conducted per variable, and the mean decrease in prediction accuracy was calculated. Variable significance was determined by the 95% confidence interval

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of permutation scores, providing a robust estimate of predictor contributions.

Causal relationships

Based on the variable importance results from the RF models, influential environmental predictors were selected to construct Partial Least Squares Structural Equation Models (PLS-SEM) for examining the direct and indirect pathways linking environmental factors to forest Rt and Rs. PLS-SEM was chosen because it can simultaneously model multiple causal pathways and latent variable relationships, and is relatively robust to non-normal data, multicollinearity, and complex multivariate structures, which are common in large-scale spatial ecological datasets (Hair et al. 2014; Ravand and Baghaei 2016).

The conceptual model included six latent variables: geography, climate IAV, freezing intensity, phenology, forest structure, and soil properties. To ensure directional consistency within the freezing intensity construct, Tminabs was reverse-transformed so that higher values represented stronger minimum-temperature stress, i.e., lower original Tminabs. The PLS-SEM analysis was conducted using the "plspm" package (v. 0.5.0) in R (Sanchez 2013). Model quality was evaluated using the coefficient of determination (R^2) and goodness-of-fit (GOF), where R^2 quantifies the explained variance of endogenous latent variables and GOF reflects the overall predictive performance of the model.

Results

Effect of EIS on forest GPP and stability

Overall, GPP responses to the two EIS events varied significantly across forest types in mid-to-north subtropical China (Fig. 2). Except for DBF, GPP declined across all other forest types, with a more pronounced reduction during the 2008 event than in 2018. Independent *t*-tests indicated that both Rt and Rs differed significantly between the two events ($p < 0.0001$; Fig. 3), and one-sample *t*-tests confirmed that all stability indices across forest types significantly deviated from the steady-state baseline of 1.0 ($p < 0.0001$; Fig. 3). A contrasting stability pattern emerged: during the more severe 2008 event, forests (except DBF) generally exhibited lower resistance (Rt < 1.0) coupled with higher resilience (Rs > 1.0). Conversely, the milder 2018 event was characterized by enhanced resistance but significantly lower resilience compared to the 2008 event. Across the four forest types, DBF consistently demonstrated superior stability, recording the highest values for both mean Rt (1.047 in 2008) and mean Rs (1.041 in 2008 and 1.005 in 2018). In contrast, the minimum resistance was observed in EBF (Rt = 0.961 in 2008), while the lowest resilience was found in MF (Rs = 0.985 in 2018).

Rt and Rs responses were analyzed along the FD gradient, as FD integrates daily temperature and precipitation and captures the temporal dynamics of freezing events. The response of forest stability to FD exhibited distinct patterns between the two events (Fig. 4). In the 2008 event, Rt for ENF, EBF, and MF showed a significant linear decline as FD prolonged ($p < 0.05$), with EBF consistently exhibiting the lowest resistance. In contrast,

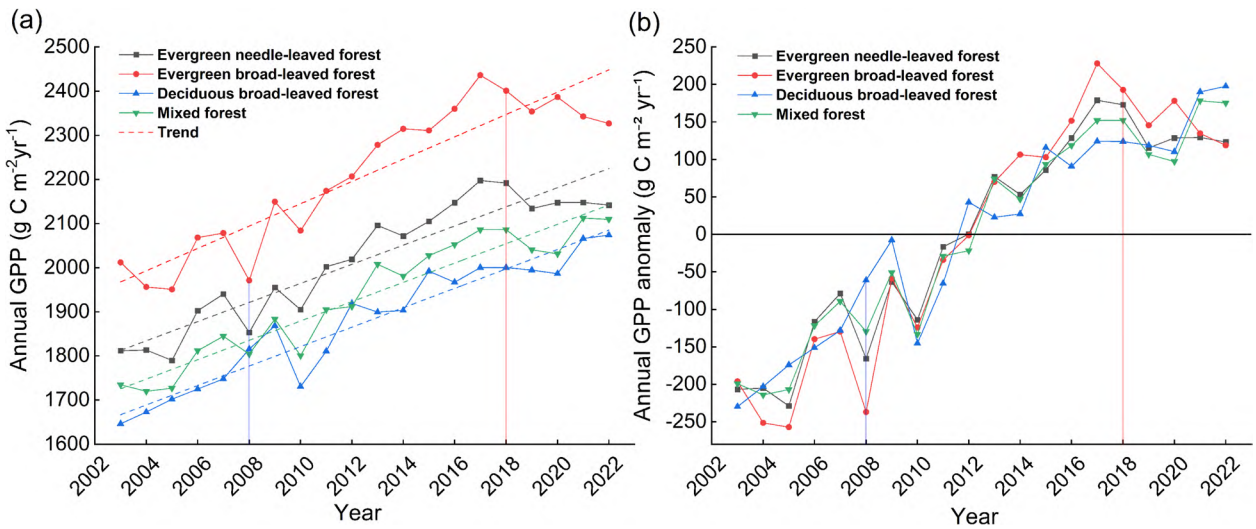


Fig. 2 **a** Annual variation and long-term trend of gross primary productivity (GPP) across the four forest types. **b** Annual GPP anomalies (2003–2022) for the four forest types. The vertical solid lines indicate the ice storm event occurred in 2008 (blue) and 2018 (red)

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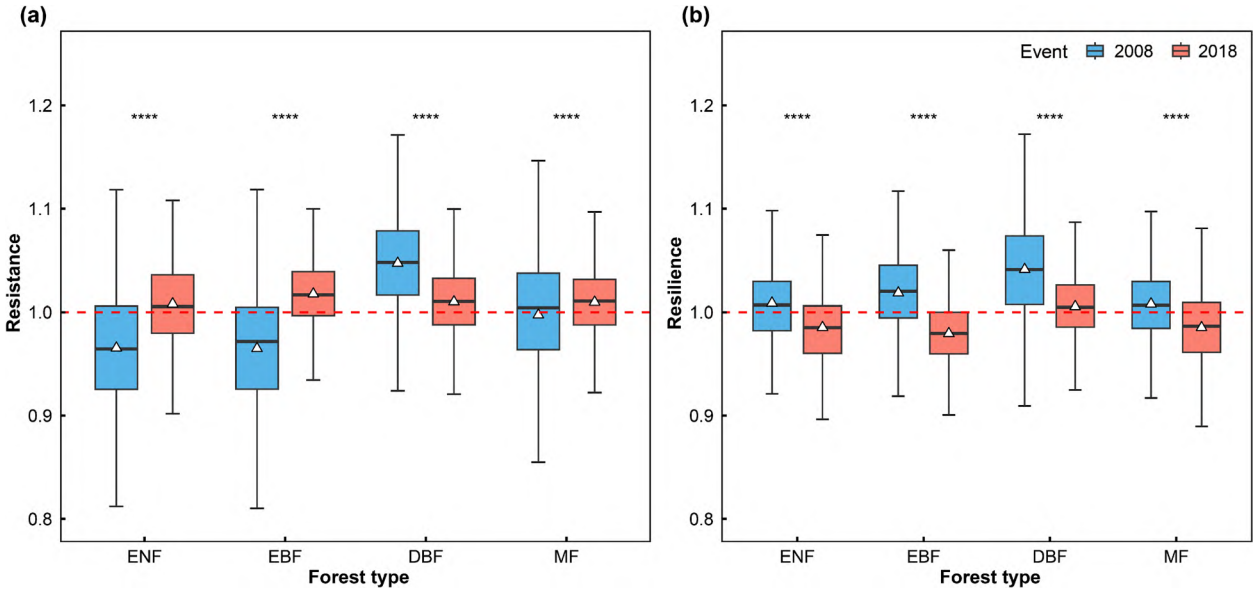


Fig. 3 Comparison of forest resistance (Rt, **a**) and resilience (Rs, **b**) during the 2008 (blue boxes) and 2018 (red boxes) ice storm events across four forest types. Box plots represent the 25th, 50th (median, horizontal lines), and 75th percentiles, with whiskers indicating the minimum and maximum values. White triangles denote the mean values for each group. The red dashed line indicates the steady-state baseline (value = 1.0). Asterisks above the boxes indicate highly significant differences between the 2008 and 2018 events for each forest type (****$p < 0.0001$, independent *t*-test). All observed Rt and Rs values significantly deviated from the baseline of 1.0 across all forest types (****$p < 0.0001$, one-sample *t*-test)

DBF followed a bell-shaped polynomial trend, maintaining Rt > 1.0 when FD ranged from 8 to 28 days (Fig. 4a). Regarding Rs, all forest types (except DBF) in 2008 exhibited a significant linear decrease ($p < 0.05$) with increasing FD, indicating that prolonged freezing significantly hindered the recovery capacity across all forest categories (Fig. 4b). In the 2018 event, the response patterns differed markedly from 2008. Both Rt and Rs across all forest types showed significant linear increases as FD increased ($p < 0.05$, Fig. 4c, d). Notably, EBF demonstrated a higher sensitivity and stronger linear correlation in Rt ($R^2 = 0.695$) and Rs ($R^2 = 0.749$) compared to other types. While most responses were linear, resistance in MF in 2018 was best characterized by a polynomial fit ($R^2 = 0.376$), showing a slight non-linear increase at higher FD levels (Fig. 4c). Overall, DBF exhibited unique stability dynamics, particularly in 2008, where it showed higher resistance at intermediate durations and a faster initial response compared to other forests.

Driving factors on forest Rt and Rs to EIS

After VIF-based screening excluded Lat and SOM, 20 environmental predictors were retained for the RF analysis. The models demonstrated robust performance, explaining 47.1%–78.2% of the variance in Rt and 44.2%–83.1% in Rs across the four forest types (Fig. 5; Supplementary Table S3). Based on the importance analysis, freezing intensity (comprising FD, Pcum, and Tminabs) emerged as the predominant determinant of forest stability in most cases. Specifically, freezing intensity accounted for 36.1% (DBF) to 61.3% (ENF) of the variation in Rt (Fig. 5a, e), and 34.5% (DBF) to 62.7% (EBF) of the variation in Rs (Fig. 5c, e). Within this category, FD and Pcum were consistently identified as the most important individual predictors (Fig. 5).

Geographical factors (e.g., longitude and elevation) represented the second most influential dimension, particularly for MF, where they contributed 25.0% to Rt and 21.7% to Rs (Fig. 5g, h). Notably, climate IAV and phenological variables showed distinct importance in EBF and DBF (Fig. 5c–f). For example, climate IAV was the leading driver of Rs in DBF (36.5%), even exceeding the contribution of freezing intensity (34.5%) (Fig. 5f). Similarly, phenological variables (particularly SOS) contributed substantially to Rt in DBF (15.7%) and EBF (12.2%), highlighting the role of biological rhythms in mediating forest responses to EIS (Fig. 5c, e). In comparison, soil properties and forest structure exhibited relatively limited impacts on both Rt and Rs across all forest types, generally accounting for less than 15% of the observed variation. Finally, most environmental predictors were highly significant ($p < 0.001$), except for a few topographic and soil variables (e.g., aspect and soil porosity) in certain forest types (Fig. 5).

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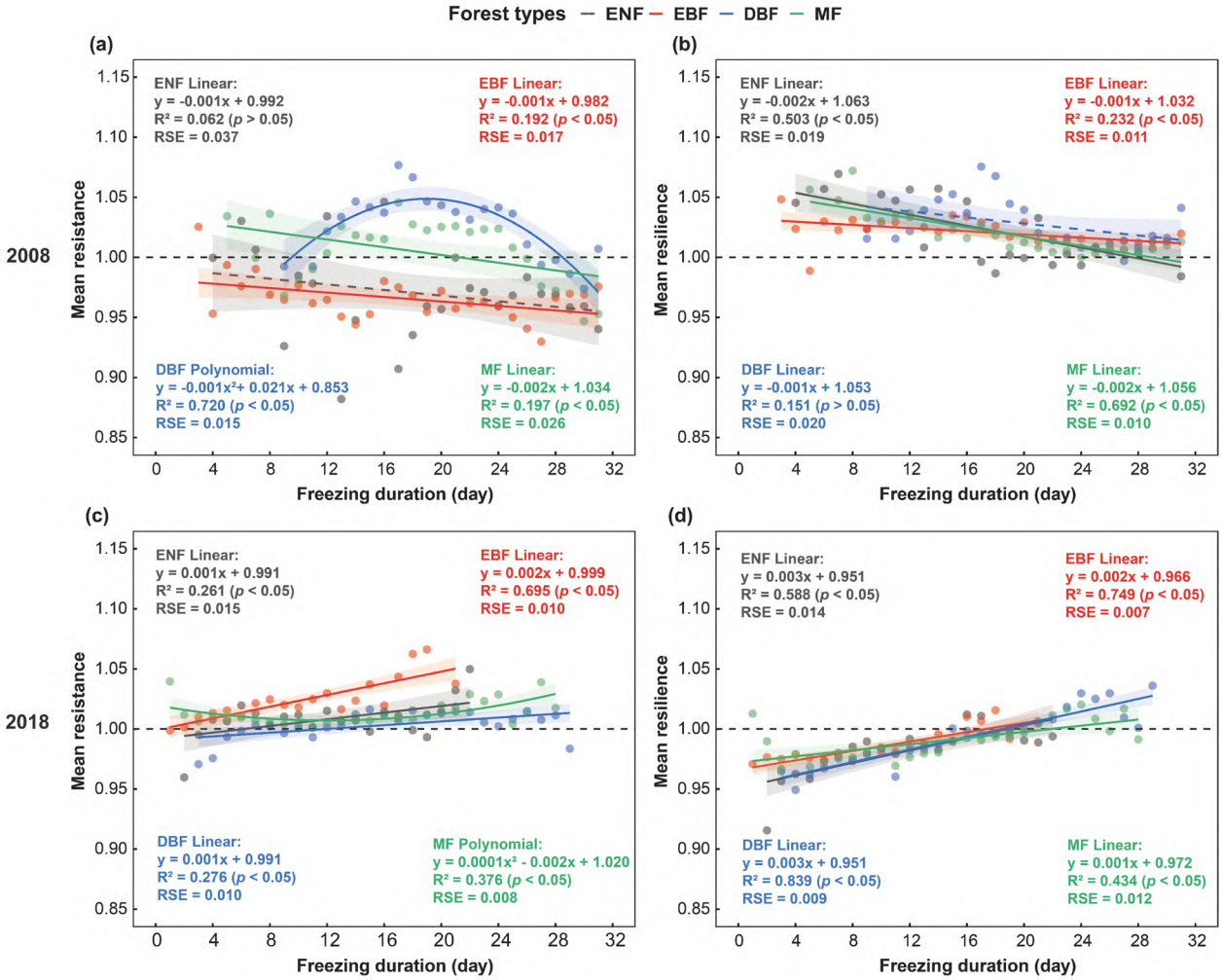


Fig. 4 Response of forest resistance (Rt) and resilience (Rs) to freezing duration during the 2008 and 2018 events. **a**, **c** Represent Rt and **b**, **d** represent Rs for the 2008 and 2018 events, respectively. Solid and dashed lines indicate significant ($p < 0.05$) and non-significant ($p > 0.05$) linear or polynomial fits. The model accuracy is evaluated by R^2 and RSE

Regulation pathway on forest Rt and Rs

Excluding the non-significant variables of slope and aspect, 18 environmental predictors were used to construct the PLS-SEM, with LAI, Pre, and POR further omitted from the EBF model. PLS-SEM results revealed that freezing intensity had significant yet divergent direct effects on Rt (Fig. 6), exerting a positive influence in MF (0.26) but strong negative impacts in EBF (−0.42) and DBF (−0.47). Phenological factors displayed a dual role, impairing Rt in ENF (−0.16) and DBF (−0.32) while enhancing it in EBF (0.09) and MF (0.16). Geographical factors generally exhibited positive direct effects on Rt in ENF, EBF, and MF (0.06 to 0.11), whereas they negatively impacted DBF (−0.20). Forest structure significantly constrained Rt in DBF (−0.13) and MF (−0.14), with a slight positive effect observed only in EBF (0.05). Additionally, climate IAV emerged as a dominant positive driver for Rt in DBF, contributing substantial direct and total effects.

(See figure on next page.)

Fig. 5 Relative importance of environmental factors in driving forest Rt and Rs based on Random Forest models. **a**, **c**, **e**, **g** and **b**, **d**, **f**, **h** Show the variable importance for Rt and Rs across the four forest types, respectively. The bar length represents the "Decrease in accuracy score," indicating the relative contribution of each predictor. Significance levels are indicated as: ns represents $p > 0.05$; * represents $p < 0.05$; ** represents $p < 0.01$; *** represents $p < 0.001$. Donut charts summarize the cumulative importance (%) of the six ecological groups. Predictor abbreviations and groups are defined in Table 1

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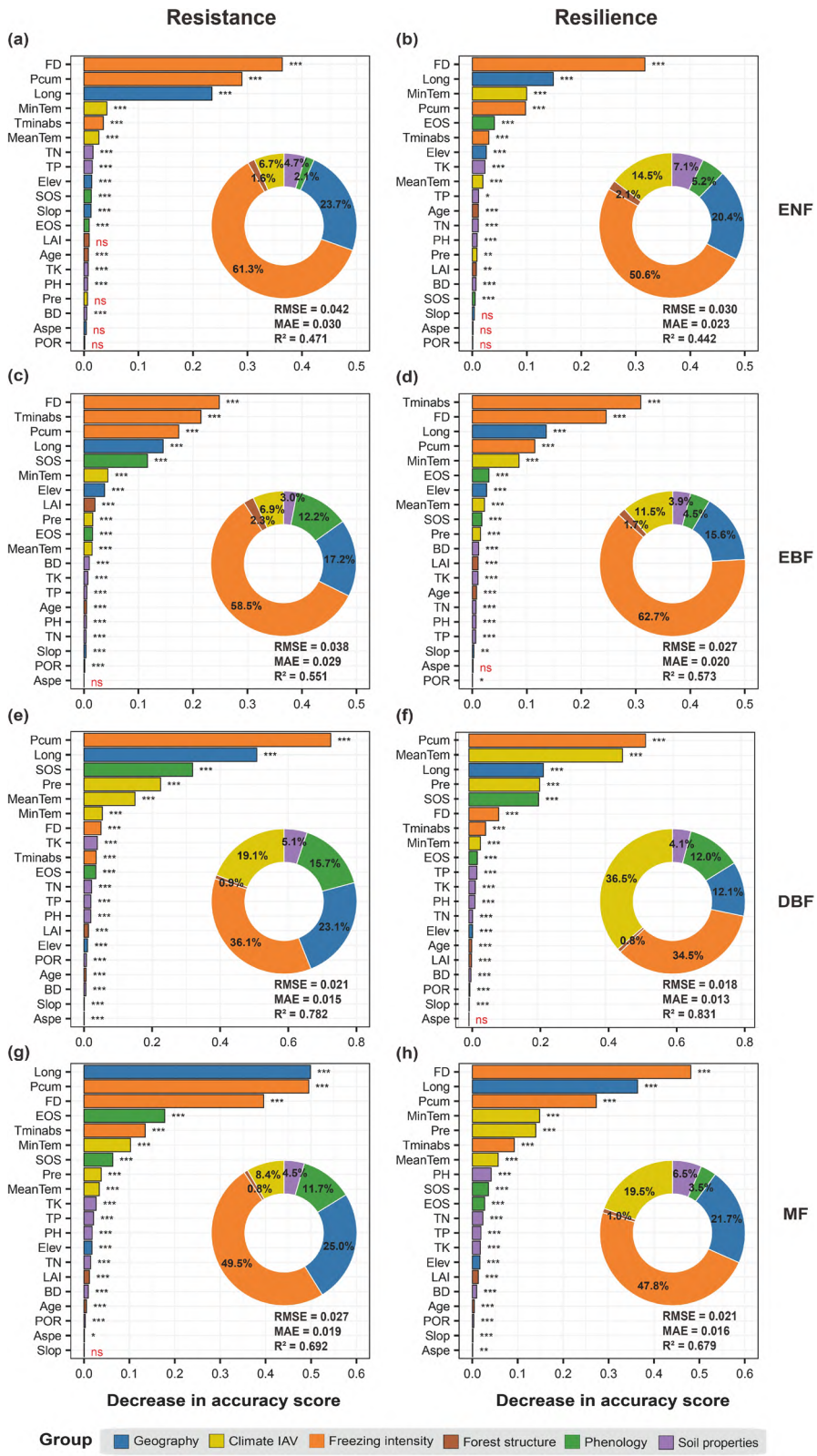


Fig. 5 (See legend on previous page.)

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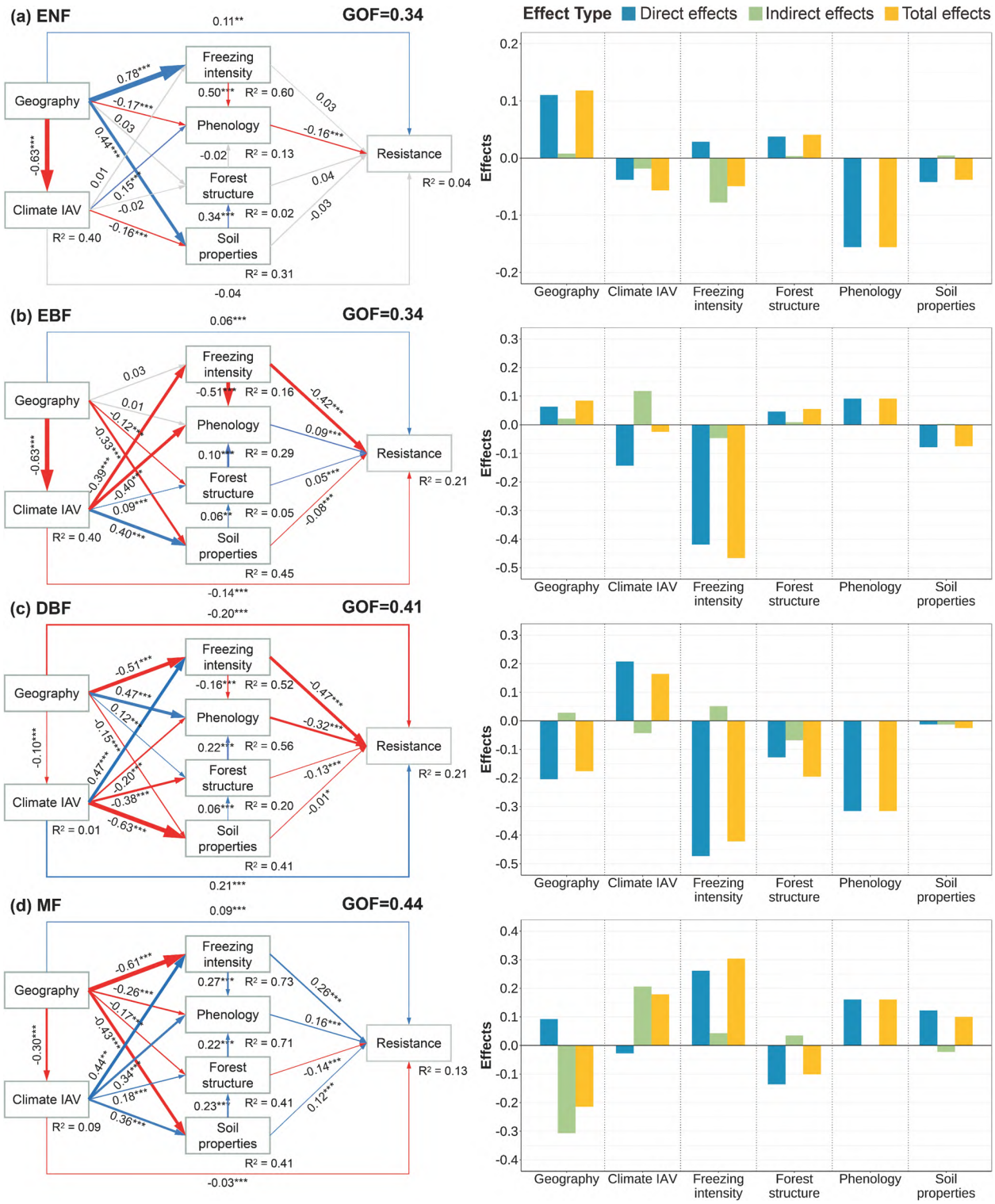


Fig. 6 The optimal path model illustrates how different environmental factors influence Rt across forest types. **a**–**d** Show the regulatory pathways of environmental variables on ENF, EBF, DBF, and MF, along with their corresponding direct, indirect, and total effects. Red arrows denote significant negative relationships, blue arrows indicate significant positive relationships, and gray arrows represent non-significant links. Arrow thickness reflects the strength of causal paths, with standardized path coefficients shown alongside each arrow. R^2 indicates the proportion of variance explained, and GOF denotes the goodness-of-fit of the entire model. Significance levels are indicated as: * represents $p < 0.05$; ** represents $p < 0.01$; *** represents $p < 0.001$. The same as below. Detailed PLS-SEM parameters for Rt are provided in Supplementary Tables S4–S8

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Regarding Rs, freezing intensity promoted it in ENF (0.26) and MF (0.18), but exerted significant negative effects in EBF (−0.33) and DBF (−0.34) (Fig. 7). Both phenology and forest structure consistently exerted negative direct effects on Rs across ENF, EBF, and DBF, whereas they functioned as positive drivers exclusively in MF (0.14 and 0.10, respectively). Geographical factors enhanced Rs in ENF (0.14) and MF (0.27) but negatively affected DBF (−0.09). Soil properties generally showed positive direct effects on Rs in ENF (0.08), EBF (0.09), and MF (0.03), with a slight negative effect found only in DBF (−0.03). Similar to Rt, climate IAV demonstrated a substantial direct positive effect on Rs specifically in DBF (0.40). The complete effect sizes, factor loadings, and other PLS-SEM parameters are provided in Supplementary Tables S4–S13.

Discussion

Divergent responses of GPP and stability across forest types

Extreme ice storms exerted pronounced yet heterogeneous impacts on forest GPP and stability across southern China. Both events reduced GPP in most forest types, particularly in evergreen forests (ENF and EBF), whereas DBF showed neutral or even positive responses (Fig. 2). This pattern aligns with previous studies showing that EIS constrains forest growth through structural damage, such as canopy breakage and tree mortality, thereby reducing carbon assimilation (King et al. 2005; Frazier et al. 2013; Klopčič et al. 2020). During the 2008 event, mechanical damage dominated over frost injury, accounting for most tree mortality in EBF and similarly in ENF (Zhang et al. 2008; Wang et al. 2016). Consequently, the higher freezing intensity in 2008 led to a more pronounced decline in GPP compared to 2018, highlighting the sensitivity of forest carbon uptake to disturbance severity (Hufkens et al. 2012). In contrast, DBF exhibited increased GPP following EIS, particularly in 2008. This counterintuitive response likely reflects distinct phenological and functional strategies. Enhanced chilling accumulation may accelerate dormancy release and advance the growing season, extending the photosynthetically active period (Gao et al. 2024; Wang et al. 2024a), while leaf shedding reduces exposure to freezing damage and facilitates rapid recovery and compensatory growth (Zanne et al. 2014; Lu et al. 2020).

Forest stability exhibited significant differences among forest types, with DBF showing the highest Rt and Rs (Fig. 3). This advantage likely reflects greater freezing tolerance and the buffering effect of winter dormancy, which reduces metabolic demand during stress and preserves carbon reserves for post-disturbance recovery (Arakawa et al. 2018; Lu et al. 2020). In contrast, evergreen forests remain physiologically active during winter, making them more susceptible to cumulative stress and structural damage.

The effects of FD on stability diverged markedly between the two events, indicating a shift in the dominant disturbance regime (Fig. 4). In 2008, prolonged FD led to a linear decline in both Rt and Rs, reflecting the combined effects of mechanical damage and physiological stress. Extreme ice loading increased canopy weight, causing structural failure (Zhou et al. 2011), while sustained freezing induced cell dehydration, membrane damage, and depletion of non-structural carbohydrate reserves, thereby constraining recovery (Zhang et al. 2016; Song et al. 2024). Such physiological stress can impair membrane integrity and photosynthetic processes, particularly in acquisitive species that are more vulnerable to functional decline under freezing disturbances (Yang et al. 2026). In contrast, the positive relationship observed in 2018 suggests that, in the absence of severe mechanical damage, prolonged freezing may confer physiological benefits. Cold acclimation became dominant, enhancing tolerance by modifying xylem properties, including increased relaxation time (τ) and elevated low-temperature thresholds (LT50) (Song et al. 2024). These contrasting responses indicate that the ecological effects of freezing depend not only on duration but also on its interaction with disturbance intensity.

Drivers and mechanisms of stability across forest types

Forest stability under ice-storm disturbance was governed by a complex interplay of climatic, geographical, and biological drivers (Guo et al. 2023). To clarify the underlying mechanisms, we applied PLS-SEM to disentangle the direct and indirect pathways through which these factors influenced forest stability. The models demonstrated acceptable explanatory power across forest types (GOF > 0.33), thereby validating the reliability of the inferred relationships (Ren et al. 2023).

Freezing intensity exerted the strongest effects on forest stability, although its impacts varied markedly across forest types. In EBF and DBF, reduced Rt was mainly associated with intensified minimum-temperature stress (lower Tminabs), lower Pcum, and longer FD, whereas reduced Rs was driven primarily by lower Tminabs. These patterns likely reflect the combined impacts of physiological stress and structural damage under freezing conditions, including cellular dehydration, membrane injury, and xylem dysfunction (Wang et al. 2020). In evergreen forests, persistent canopy foliage may further amplify ice loading and cumulative stress, thereby heightening vulnerability to disturbance (Song et al. 2024). By contrast, seasonal leaf shedding in DBF likely mitigates direct exposure to freezing injury, while sufficient precipitation

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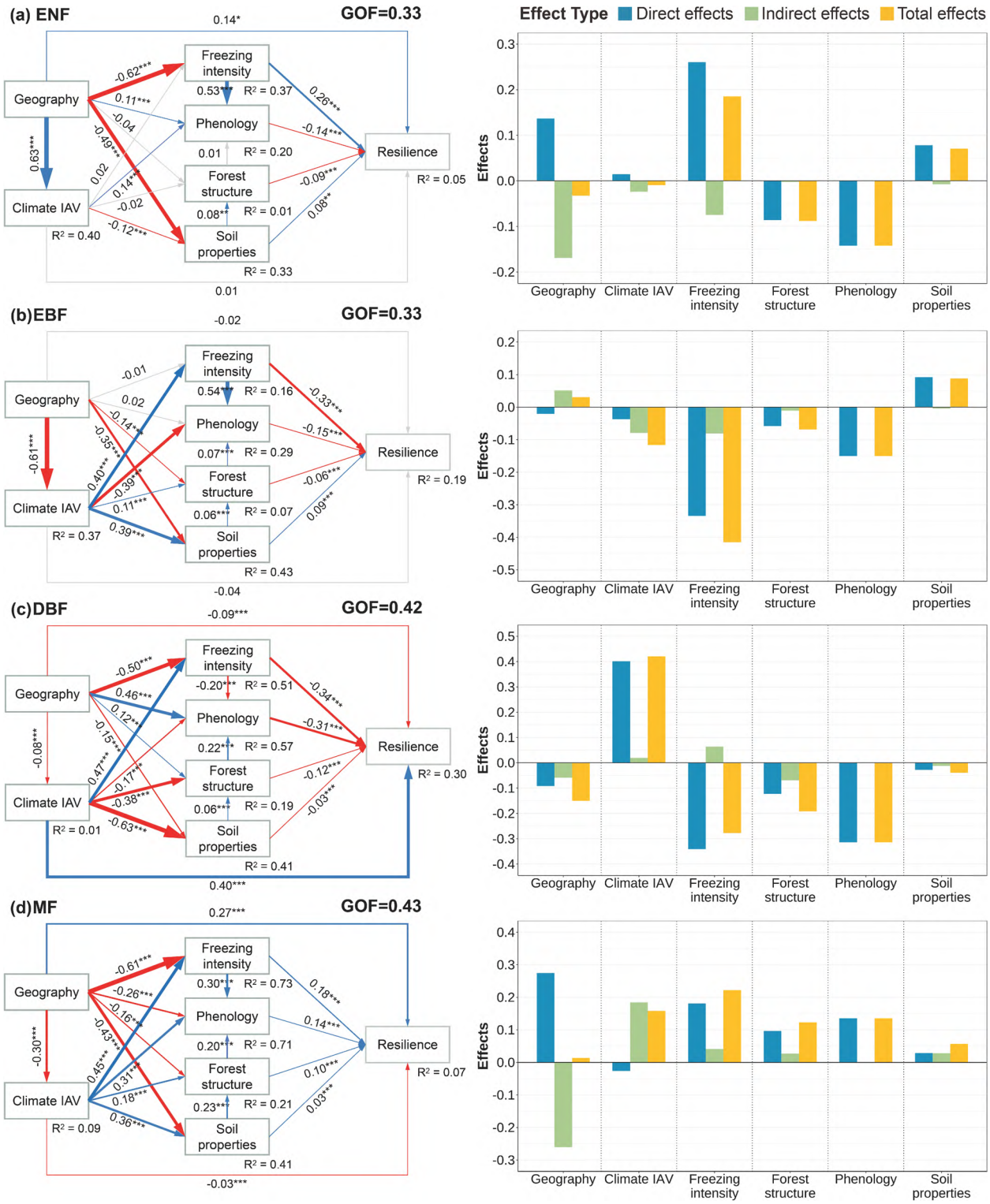


Fig. 7 The optimal path model illustrates how different environmental factors influence Rs across forest types. **a**–**d** Represent ENF, EBF, DBF, and MF, respectively. Detailed PLS-SEM parameters for Rs are provided in Supplementary Tables S9–S13

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may help alleviate water limitation and support post-disturbance recovery (Xie et al. 2015; Seyednasrollah et al. 2020). In contrast to other types, MF exhibited higher stability under lower Tminabs and lower Pcum, suggesting a greater tolerance to cold but relatively dry freezing conditions. Lower temperatures may selectively remove frost-sensitive individuals and promote community reassembly, while reduced precipitation likely reduces ice accretion and canopy loading, thereby averting mechanical damage. Collectively, these processes may facilitate structural reorganization and enhance ecosystem-level stability in MF (Seidl et al. 2014).

Geographical factors mainly influenced forest stability indirectly by shaping the climatic background of freezing disturbance. Based on the results of RF and PLS-SEM, their stronger effect in MF indicates its heightened responsiveness to broad-scale environmental gradients. The greater stability observed at higher elevations and lower longitudes may reflect the strong spatial heterogeneity created by the large mountain systems in the western study region. These mountains likely increase spatial heterogeneity and create local refugia and microclimatic buffering, thereby providing favorable conditions for the stable persistence of mixed forests (Suggitt et al. 2018). Climate IAV further played a dual role, both amplifying the effects of freezing intensity and exerting an independent influence on Rt and Rs, suggesting that long-term climatic variability contributes to ecosystem sensitivity and adaptive capacity (Anderson et al. 2011). Notably, phenology exerted the strongest negative effect on Rt and Rs in DBF (total effect > 0.3), with higher Rt and Rs associated with delayed SOS and earlier EOS. This demonstrates that a shorter active canopy period may enhance stability under freezing disturbance by reducing the exposure of vulnerable tissues to frost damage, as delayed spring leaf-out lowers the risk of late frost, whereas earlier autumn senescence promotes cold acclimation and limits freezing exposure. Such a phenological pattern is consistent with frost avoidance in deciduous woody plants (Vitasse et al. 2014; Zanne et al. 2014).

Furthermore, the effects of forest structure on stability exhibited distinct patterns across forest types. The strongest negative total effect was observed in DBF (approximately −0.2), where stability declined mainly with increasing LAI, consistent with previous findings (Xiang et al. 2025). Conversely, for ENF and EBF, stability was predominantly governed by stand age, with older forests exhibiting a pronounced trade-off between higher Rt and lower Rs. This indicates that while the structural complexity and biomass of mature evergreen stands enhance their resistance to initial impacts, their resilience is hampered by the metabolic costs of maintaining large biomass and the inherently slower growth of aging trees (Wang et al. 2022). Additionally, stability in MF was primarily driven by LAI, with denser canopies promoting Rt at the expense of Rs. Such structural density bolsters short-term resistance through microclimatic buffering and sustained photosynthetic capacity; however, the associated metabolic costs and intensified competition likely constrain the recovery phase, thereby limiting overall resilience (Liu et al. 2024).

Implications for forest management and sustainability under climate extremes

The heterogeneous responses of forest stability to EIS have important implications for regional carbon markets and socio-economics. In evergreen forests, EIS may disrupt carbon balance by suppressing GPP and potentially increasing post-disturbance carbon release, underscoring the vulnerability of evergreen-dominated sinks, amplifying uncertainties in carbon accounting, and cascading into risks for ecosystem service provision and regional economic stability (Zhou et al. 2011; Li et al. 2021b). These findings necessitate a management shift from productivity orientation toward integrated risk-resilience strategies (Seidl et al. 2016). Given that canopy structural complexity bolsters stability, targeted silviculture—including scientific thinning to reduce canopy density and ice-loading risk, and the transition to age-structured, mixed-species systems—offers a robust pathway to enhance resistance in high-risk monocultures (Liu et al. 2024; del Río et al. 2022). While DBF and MF exhibit greater stability, this reflects a fundamental trade-off between sequestration efficiency and disturbance tolerance rather than a direct substitute for evergreen systems (Baker et al. 2023). Consequently, instead of uniform strategies, climate-adaptive planning should leverage spatial heterogeneity and functional complementarity by prioritizing vulnerable evergreen stands in high-risk freezing regions and retaining DBF and MF as resilience-supporting components within landscape mosaics. Specifically, in areas with high FD and low Tminabs, protection and structural improvement should be prioritized for vulnerable EBF and ENF stands; in restoration and afforestation practices, the proportion of DBF or MF components could be moderately increased to enhance landscape-level resilience; and in evergreen forests with high LAI, light to moderate thinning may help reduce canopy ice loading and mechanical damage risk. Strategically configuring evergreen, deciduous, and mixed stands can effectively reconcile carbon sequestration with stability, securing regional carbon sinks against intensifying climate extremes.

The acquisition of high-resolution, near-real-time, and robust spatial data has become a central driver of modern forest governance and socio-economic decision-making.

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Accurate identification of forest dynamics under extreme climate disturbances is essential for advancing data-driven governance and precision forestry. As a direct proxy of photosynthetic activity, SIF is tightly coupled with canopy-scale carbon assimilation processes, reducing the response lag of conventional greenness indices under extreme stress conditions and improving the early detection and real-time monitoring of forest carbon sink variability (Hao et al. 2021; Shekhar et al. 2022). By integrating machine learning algorithms, the SIF-GPP-based monitoring framework enables continuous tracking of forest degradation and post-disturbance recovery trajectories, while supporting more efficient resource allocation for harvesting, thinning, and restoration strategies. Such predictive frameworks have demonstrated broad applicability across carbon markets, energy systems, agricultural resource governance, and circular economy models (Zhang and Xu 2021; Xu and Zhang 2022; Jin and Xu 2025, 2026), highlighting the potential of technological integration to enhance social welfare and environmental sustainability. Importantly, this framework provides a robust decision-support basis for forest vulnerability assessment and precision governance under climate extremes.

Nevertheless, several uncertainties should be acknowledged when applying SIF-derived GPP to extreme ice storm events. Persistent cloud cover and snow/ice conditions during January–February may affect satellite observation quality, although the 8-day compositing strategy can partly reduce short-term atmospheric noise. Meanwhile, this temporal resolution may smooth abrupt physiological declines immediately after ice storms; therefore, our results should be interpreted as event-scale and annual functional responses rather than instantaneous canopy damage. In addition, the conversion from SIF to GPP relies on empirical and model-based assumptions, and stress-induced changes in the fluorescence–photosynthesis relationship under extreme cold conditions may introduce additional uncertainty (Yang et al. 2021). Future studies integrating SIF-GPP with flux-tower observations, high-resolution optical imagery, and field-based damage surveys would further improve the validation of forest responses to extreme ice storms.

Conclusions

This study disentangles the multifaceted impacts of EIS on forest productivity and stability in mid-to-north subtropical China. While EIS generally reduced GPP—with the severe 2008 event exerting a more pronounced suppressive effect—DBF exhibited superior resistance and resilience compared to the more vulnerable EBF. Notably, the 2008 event was characterized by lower Rt but higher Rs, whereas the opposite occurred in 2018. Although freezing intensity is the primary driver of stability, its influence is significantly modulated by climatic, geographical, and biological factors through forest-type-specific pathways. By bridging the knowledge gap in subtropical ecosystems, we elucidate the nexus between forest management and socio-economic stability under EIS. Our results advocate for a transition toward adaptive governance—one that leverages forest-type heterogeneity, spatial risk differentiation, and functional complementarity to bolster ecosystem stability against extreme climate.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13717-026-00705-5.

Supplementary material 1.

Author contributions

Yipei Zhao: Writing—original draft, methodology, software. Qi Wang: Methodology. Wen Nie: Validation. Shaowei Yang: Methodology. Xiangfen Cheng: Software. Yuanwen Kuang: Writing—review and editing, resources. Jianfeng Liu: Writing—review and editing, supervision, funding acquisition.

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Availability of data and materials

Data will be made available on request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing financial interests.

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